

A Course Based Project Report on
Virtual Disk Manager

Submitted to the
Department of CSE - (CyS, DS) and AI&DS

in partial fulfilment of the requirements for the completion of course
OPERATING SYSTEMS (22PC11T202)

BACHELOR OF TECHNOLOGY

IN

CSE-DATA SCIENCE

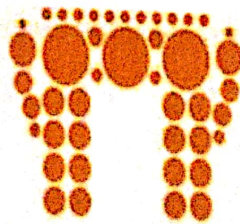
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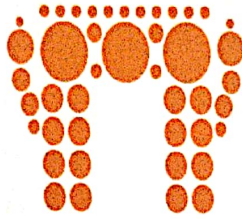
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CERTIFICATE

This is to certify that the project report entitled “**Virtual Disk Manager**” is a bonafide work done under our supervision and is being submitted by **Mr. B. Vamsi Kiran (24071A6708)**, **Mr. B. Bhuvanesh Gupta (24071A6709)**, **Mr. G. Sanath Kumar (24071A6714)**, **Mr. G. Shreeman (24071A6720)**, **Mr. A. Sai Vignesh (25075A6701)** in partial fulfillment for the award of the degree of **Bachelor of Technology in CSE-Data Science**, of the VNRVJIET, Hyderabad during the academic year 2025-2026.

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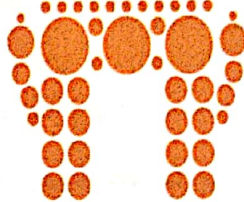
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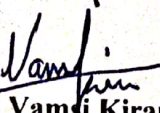


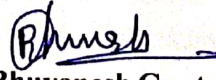
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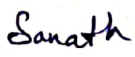
DECLARATION

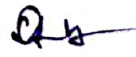
We declare that the course based project work entitled “**Virtual Disk Manager**” submitted in the Department of **CSE-(CyS, DS) and AI&DS**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in CSE-Data Science** is a bonafide record of our own work carried out under the supervision of **Dr. N. Madhuri, Assistant Professor, Department of CSE-(CyS, DS) and AI&DS, VNRVJIE**T. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

Disk management is a crucial component of modern operating systems, as it directly influences the efficiency of data storage and retrieval processes. Since disk access operations are significantly slower compared to CPU operations, optimizing disk performance is essential for improving overall system throughput. Two major aspects that affect disk performance are disk scheduling and file allocation. Disk scheduling determines the order in which disk I/O requests are serviced, while file allocation defines how data is organized and stored on disk blocks. This project presents the design and implementation of a **Virtual Disk Manager with Disk Scheduling and File Allocation Simulation**, which serves as an educational tool to demonstrate and analyze different disk management strategies. The system supports multiple disk scheduling algorithms, including First Come First Serve (FCFS), Shortest Seek Time First (SSTF), SCAN, C-SCAN, and C-LOOK. These algorithms are implemented to simulate the movement of the disk head across cylinders and calculate the total seek time required to service a sequence of disk requests. In addition to disk scheduling, the project also implements file allocation techniques such as Sequential Allocation, Linked Allocation, and Indexed Allocation. These techniques illustrate how files are stored and accessed in secondary storage, and how different methods impact fragmentation and access efficiency. The system is developed using JavaScript, with a React-based frontend that provides an interactive interface for users to input disk requests and visualize results. The project also includes graphical representations of disk head movement, allowing users to clearly understand the behavior of each algorithm. Furthermore, a comparative analysis feature is included to evaluate the performance of different algorithms under the same conditions. Overall, this project bridges the gap between theoretical concepts and practical implementation in operating systems, providing a clear understanding of disk scheduling and file allocation mechanisms.

CHAPTER-1

INTRODUCTION

Disk management is one of the most important responsibilities of an operating system, as it deals with the storage and retrieval of data from secondary storage devices such as hard disk drives (HDDs). A hard disk is a magnetic storage device that consists of one or more rotating platters coated with magnetic material. Data is stored in the form of tracks and sectors on these platters, and a read/write head is used to access the data. The movement of this head across different tracks introduces a delay known as seek time, which significantly affects the performance of the system.

In a multi-programming environment, multiple processes generate disk I/O requests simultaneously. These requests are stored in a queue, and the operating system must decide the order in which they should be serviced. If the requests are handled inefficiently, the disk head may move back and forth unnecessarily, resulting in increased seek time and reduced system performance. Therefore, efficient disk scheduling algorithms are required to optimize head movement and improve throughput.

Several disk scheduling algorithms have been developed to address this problem. The First Come First Serve (FCFS) algorithm processes requests in the order they arrive, making it simple but inefficient in many cases. The Shortest Seek Time First (SSTF) algorithm selects the request closest to the current head position, reducing seek time but potentially causing starvation. The SCAN algorithm, also known as the elevator algorithm, moves the disk head in one direction servicing requests and then reverses direction. C-SCAN and C-LOOK further optimize this approach by providing more uniform wait times.

Another important aspect of disk management is file allocation. File allocation methods determine how data blocks are stored on disk. Sequential allocation stores files in contiguous blocks, making access faster but causing fragmentation. Linked allocation connects blocks using pointers, eliminating fragmentation but increasing access time. Indexed allocation uses an index block to store pointers, providing efficient access without fragmentation.

Understanding these concepts through theory alone can be challenging. Therefore, this project introduces a Virtual Disk Manager that simulates disk scheduling algorithms and file allocation techniques. The system provides graphical visualization of disk head movement and allows users to compare different algorithms based on performance metrics such as total seek time.

This project is designed to help students gain practical knowledge of operating system concepts by interacting with a real-time simulation environment. It demonstrates how different algorithms behave under various conditions and highlights the importance of efficient disk management in improving system performance.

CHAPTER-2

Method

The proposed system follows a structured and modular approach to design and implement a Virtual Disk Manager that simulates disk scheduling algorithms and file allocation techniques. The methodology consists of several stages, including problem analysis, algorithm design, implementation, visualization, and testing.

1. Problem Analysis and System Design

The first step involved understanding the problem of disk scheduling and file allocation in operating systems. The objective was to design a system that can simulate different algorithms and compare their performance based on seek time and efficiency. The system is divided into three main components: user interface, algorithm processing, and visualization.

2. Disk Scheduling Algorithm Implementation

The core functionality of the system is implemented using JavaScript. The following disk scheduling algorithms are included:

- **FCFS:** Processes requests in the order they arrive. It is simple but may result in high seek time.
- **SSTF:** Selects the nearest request to the current head position, reducing seek time but may cause starvation.
- **SCAN:** Moves the head in one direction servicing requests, then reverses direction.
- **C-SCAN:** Services requests in one direction and jumps back to the beginning.
- **C-LOOK:** Similar to C-SCAN but only travels up to the last request.

Each algorithm:

- Accepts input requests and head position
- Calculates total seek time

- Tracks head movement step-by-step
 - Stores results for visualization
-

3. File Allocation Techniques

The system also implements file allocation methods:

- **Sequential Allocation:** Files stored in continuous blocks
- **Linked Allocation:** Blocks connected using pointers
- **Indexed Allocation:** Uses index block for efficient access

These techniques help demonstrate storage organization and fragmentation issues.

4. Frontend Development (React)

The user interface is developed using React.js to provide an interactive and dynamic experience. The frontend allows users to:

- Enter disk requests
 - Select algorithms
 - View results instantly
 - Compare algorithms
-

5. Visualization

Graphical visualization is an important feature of the system:

- X-axis → Cylinder positions
- Y-axis → Execution order
- Lines represent head movement

This helps users easily understand algorithm behavior.

6. Comparative Analysis

All algorithms are executed for the same input and compared based on:

- Total seek time
 - Efficiency
 - Fairness
-

7. Testing and Validation

The system is tested with multiple inputs to ensure correctness. Outputs are validated with expected theoretical results.

CHAPTER-3

TEST CASES/ OUTPUT

Test Case 1

Input:

- Disk Requests: 98, 183, 37, 122, 14, 124, 65, 67
- Initial Head Position: 53
- Algorithm: FCFS

Expected Output:

- Total Seek Time: High

Actual Output:

- Total Seek Time: High

Observation:

- FCFS processes requests in arrival order
 - Causes large head movement
 - Not efficient for optimization
-

Test Case 2

Input:

- Disk Requests: Same as above
- Algorithm: SSTF

Expected Output:

- Reduced seek time

Actual Output:

- Reduced seek time

Observation:

- Selects nearest request
 - Improves performance
 - May cause starvation
-

Test Case 3

Input:

- Disk Requests: Same
- Algorithm: SCAN / C-LOOK

Expected Output:

- Balanced performance

Actual Output:

- Balanced seek time

Observation:

- Provides fair servicing
- Avoids starvation
- Suitable for real systems

Graph Output

- Head movement graphs displayed
- Clear comparison between algorithms
- Helps identify best algorithm

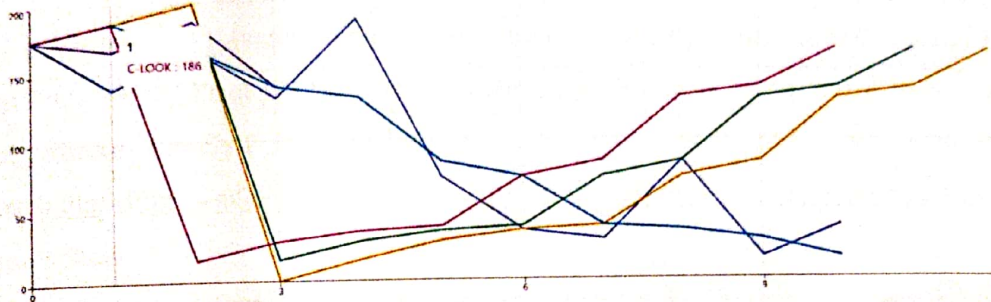
Output:



- SCAN
- C-SCAN
- FCFS

399	42.54
389	37.42
463	45.38

Head Movement Comparison



File Allocation Simulator

Visualize how files are allocated on disk using different strategies.

Disk Setup

Total Blocks

32

Reset Disk

Create File

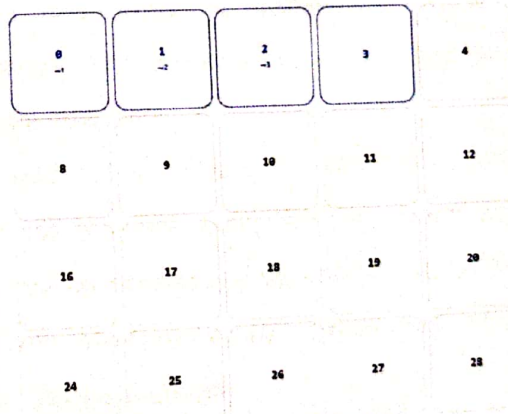
File name

4

Linked

+ Allocate

Disk Blocks



CHAPTER-4

RESULTS

The Virtual Disk Manager system was tested with multiple disk request sequences and different initial head positions to evaluate the performance of various disk scheduling algorithms, namely FCFS, SSTF, SCAN, C-SCAN, and C-LOOK. The performance of each algorithm was analyzed based on the total seek time and the pattern of disk head movement.

From the experimental results, it is observed that the **FCFS (First Come First Serve)** algorithm results in the highest seek time among all the algorithms. This is because it processes requests strictly in the order of arrival without considering the current position of the disk head. As a result, the disk head moves unnecessarily across large distances, leading to inefficient performance and increased access time.

The **SSTF (Shortest Seek Time First)** algorithm significantly reduces the total seek time by selecting the request closest to the current head position. This minimizes the movement of the disk head and improves overall performance. However, it was observed that SSTF may lead to starvation, where requests that are far from the current head position may remain unserved for a long time.

The **SCAN (Elevator Algorithm)** provides a balanced approach by moving the disk head in one direction and servicing all requests along the path before reversing direction. This reduces unnecessary head movement and ensures that all requests are eventually served. The performance of SCAN is more stable compared to SSTF, as it avoids starvation and provides fairness among requests.

The **C-SCAN (Circular SCAN)** algorithm further improves fairness by servicing requests in only one direction and then jumping back to the beginning of the disk without servicing requests in the reverse direction. This results in more uniform wait times for all requests, making it suitable for systems with heavy workloads.

The **C-LOOK** algorithm optimizes C-SCAN by moving the disk head only up to the last request in the queue before jumping back to the first request. This reduces unnecessary head movement to unused disk areas and improves efficiency compared to SCAN and C-SCAN.

In addition to disk scheduling, the project also evaluated file allocation methods. Sequential allocation was found to be simple and fast but prone to fragmentation. Linked allocation eliminated fragmentation but increased access time due to pointer traversal. Indexed allocation provided efficient direct access and reduced fragmentation, making it the most balanced approach among the three.

The graphical visualization of disk head movement clearly illustrates the differences in algorithm behavior. The plots show how algorithms like SSTF and C-LOOK reduce total head movement compared to FCFS. These visual representations make it easier to analyze and compare performance.

Overall, the results confirm that disk scheduling algorithms play a crucial role in optimizing system performance. Algorithms such as SSTF and C-LOOK provide better efficiency, while SCAN ensures fairness. The outputs obtained from the system match the expected theoretical behavior, thereby validating the correctness and effectiveness of the Virtual Disk Manager.

CHAPTER 5

Summary, Conclusion, Recommendation

5.1. Summary

This project focused on the design and implementation of a Virtual Disk Manager to simulate disk scheduling algorithms and file allocation techniques in operating systems. The primary objective of the project was to understand how disk I/O operations are managed and how different scheduling strategies affect system performance. Since disk access time is significantly higher compared to CPU execution time, optimizing disk operations becomes essential for improving overall system efficiency.

The system successfully implements multiple disk scheduling algorithms, including FCFS, SSTF, SCAN, C-SCAN, and C-LOOK. These algorithms were used to simulate disk head movement for a given set of disk requests and initial head position. The total seek time for each algorithm was calculated, and their performance was compared. In addition to disk scheduling, the project also demonstrated file allocation methods such as Sequential, Linked, and Indexed allocation, helping to understand how data is stored and accessed on disk.

The project was developed using JavaScript with a React-based frontend, providing an interactive user interface. Users can input disk requests, select algorithms, and visualize the results through graphical representations of disk head movement. The inclusion of visualization and comparative analysis makes the system an effective learning tool for operating system concepts.

5.2. Conclusion

The results obtained from the Virtual Disk Manager clearly demonstrate that disk scheduling algorithms have a significant impact on system performance. The FCFS algorithm, although simple to implement, results in higher seek time due to inefficient head movement. The SSTF algorithm improves performance by selecting the nearest request, but it may lead to starvation in certain cases. SCAN and C-SCAN provide a more balanced approach by ensuring fairness and reducing starvation. Among all the

algorithms. C-LOOK offers better optimization by avoiding unnecessary head movement and providing efficient performance.

Similarly, file allocation techniques also play an important role in disk management. Sequential allocation is simple but suffers from fragmentation, while linked allocation avoids fragmentation but increases access time. Indexed allocation provides a balanced solution with efficient access and minimal fragmentation.

Overall, the project successfully achieves its objective of simulating and analyzing disk scheduling and file allocation techniques. The outputs obtained are consistent with theoretical expectations, and the visualization component enhances understanding of algorithm behavior. This project effectively bridges the gap between theoretical knowledge and practical implementation in operating systems.

5.3. Recommendations

To further enhance the functionality and usability of the Virtual Disk Manager, the following improvements can be considered:

System Enhancements:

- Implement priority-based disk scheduling to handle real-time requests
- Add support for dynamic request generation to simulate real-world scenarios
- Improve the graphical interface with animations for better visualization

Feature Enhancements:

- Include additional performance metrics such as average seek time and wait time
- Provide detailed comparison charts for better analysis
- Allow users to export results for documentation purposes

Future Scope:

- Extend the project to include SSD scheduling techniques
- Integrate other operating system modules such as CPU scheduling and memory management
- Develop a mobile or cloud-based version for wider accessibility
- Incorporate real-time simulation with multiple user inputs

With these enhancements, the Virtual Disk Manager can be developed into a comprehensive educational tool for understanding advanced operating system concepts.

REFERENCES

1. A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 10th ed., Wiley, 2018.
2. A. S. Tanenbaum and H. Bos, *Modern Operating Systems*, 4th ed., Pearson, 2015.
3. GeeksforGeeks, "Disk Scheduling Algorithms in Operating System," [Online]. Available: <https://www.geeksforgeeks.org/disk-scheduling-algorithms/>
4. TutorialsPoint, "Operating System - Disk Scheduling," [Online]. Available: https://www.tutorialspoint.com/operating_system/os_disk_scheduling.htm
5. Operating System Concepts A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 10th ed., Wiley, 2018.
6. Modern Operating Systems A. S. Tanenbaum and H. Bos, *Modern Operating Systems*, 4th ed., Pearson, 2015.
7. Operating Systems Internals and Design Principles W. Stallings, *Operating Systems: Internals and Design Principles*, 9th ed., Pearson, 2018.
8. TutorialsPoint TutorialsPoint, "Operating System - Disk Scheduling." [Online]. Available: https://www.tutorialspoint.com/operating_system/os_disk_scheduling.htm
9. Linux Documentation Project The Linux Documentation Project, "File System Hierarchy and Disk Management."
10. The Design and Implementation of the FreeBSD Operating System M. K. McKusick et al., *The Design and Implementation of the FreeBSD Operating System*, 2nd ed., Addison-Wesley, 2014.